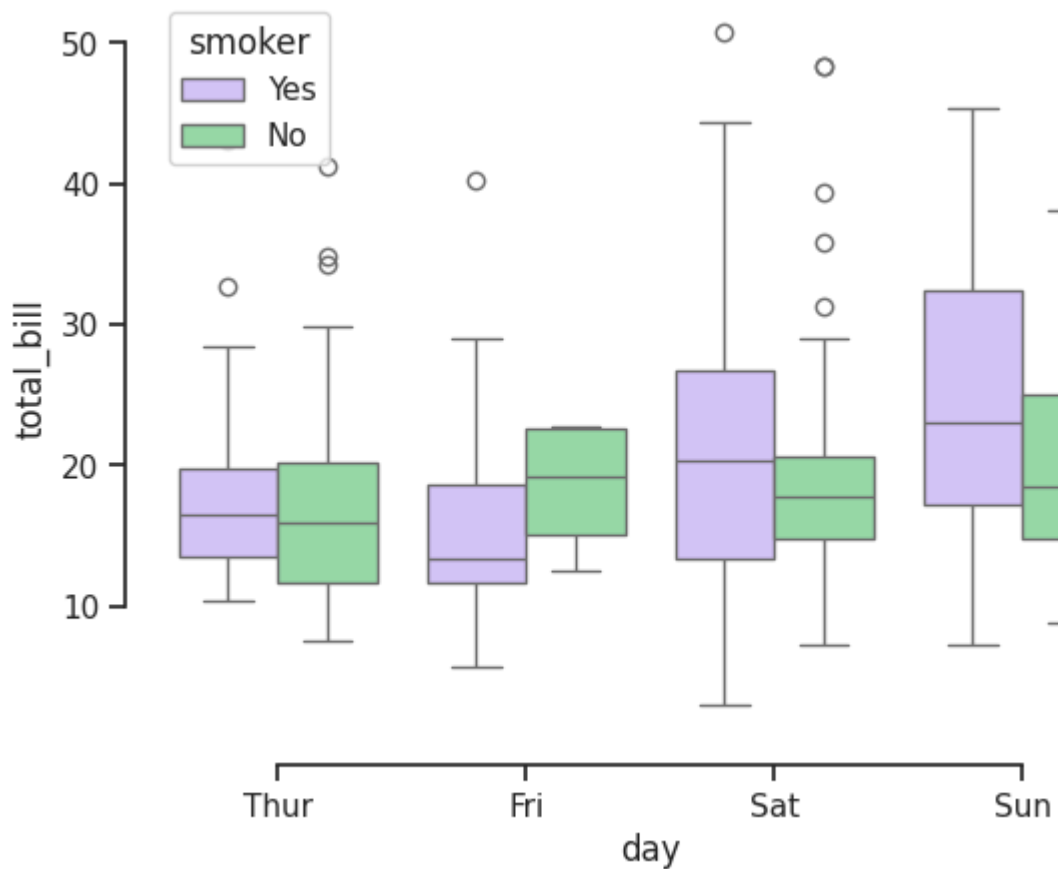


```
import seaborn as sns
sns.set_theme(style="ticks", palette="pastel")

# Load the example tips dataset
tips = sns.load_dataset("tips")

# Draw a nested boxplot to show bills by day and time
sns.boxplot(x="day", y="total_bill",
            hue="smoker", palette=["m", "g"],
            data=tips)
sns.despine(offset=10, trim=True)
```



✓ EDA(Exploratory Data Analysis)

tips

	total_bill	tip	sex	smoker	day	time	size	
0	16.99	1.01	Female	No	Sun	Dinner	2	
1	10.34	1.66	Male	No	Sun	Dinner	3	
2	21.01	3.50	Male	No	Sun	Dinner	3	
3	23.68	3.31	Male	No	Sun	Dinner	2	
4	24.59	3.61	Female	No	Sun	Dinner	4	
...	...	...	...	...	...	...	...	
239	29.03	5.92	Male	No	Sat	Dinner	3	
240	27.18	2.00	Female	Yes	Sat	Dinner	2	
241	22.67	2.00	Male	Yes	Sat	Dinner	2	
242	17.82	1.75	Male	No	Sat	Dinner	2	
243	18.78	3.00	Female	No	Thur	Dinner	2	

244 rows × 7 columns

다음 단계:

[tips 변수로 코드 생성](#)[New interactive sheet](#)

tips.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   total_bill  244 non-null   float64
1   tip         244 non-null   float64
2   sex         244 non-null   category
3   smoker      244 non-null   category
4   day         244 non-null   category
5   time        244 non-null   category
6   size        244 non-null   int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

```
df=tips
# df=tips.copy()
```

```
#df=tips
df=tips.copy()
```

✓ Univariate Analysis: Categorical Data

Let's visualize the distribution of our categorical features: `sex`, `smoker`, `day`, and `time` using count plots.

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(15, 5))

plt.subplot(1, 4, 1)
sns.countplot(x='sex', data=df, palette='viridis')
plt.title('Distribution of Sex')

plt.subplot(1, 4, 2)
sns.countplot(x='smoker', data=df, palette='viridis')
plt.title('Distribution of Smoker')

plt.subplot(1, 4, 3)
sns.countplot(x='day', data=df, palette='viridis')
plt.title('Distribution of Day')

plt.subplot(1, 4, 4)
sns.countplot(x='time', data=df, palette='viridis')
plt.title('Distribution of Time')

plt.tight_layout()
plt.show()
```

```

-----
TypeError                                Traceback (most recent call last)
/tmp/ipykernel_2528/3632114261.py in <cell line: 0>()
      5
      6 plt.subplot(1, 4, 1)
----> 7 sns.countplot(x='sex', data=df, palette='viridis')
      8 plt.title('Distribution of Sex')
      9

```

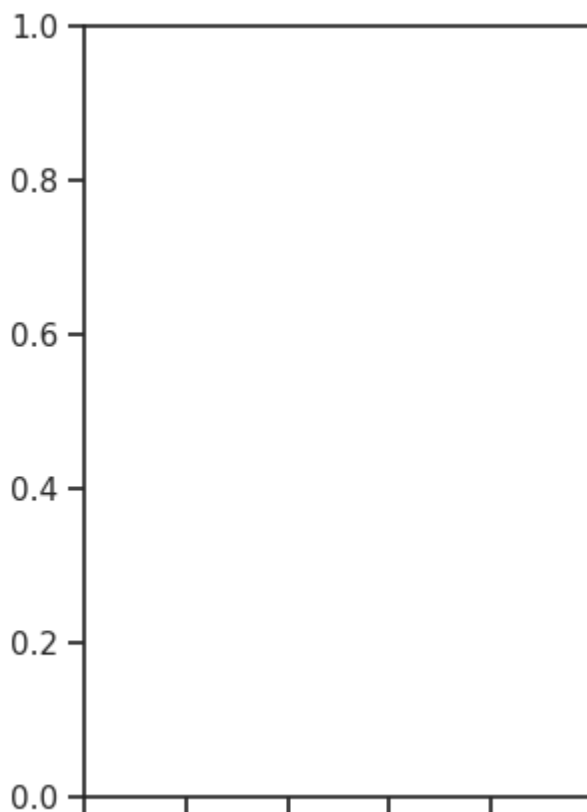
5 frames

```

/usr/local/lib/python3.13/dist-packages/seaborn/_core/data.py in
handle_data_source(data)
    276     elif data is not None and not isinstance(data, Mapping):
    277         err = f"Data source must be a DataFrame or Mapping, not
{type(data)!r}."
--> 278         raise TypeError(err)
    279
    280     return data

```

TypeError: Data source must be a DataFrame or Mapping, not <class 'str'>.



다음 단계: 오류 설명

✓ Univariate Analysis: Numerical Data

Now, let's look at the distributions of numerical features: `total_bill`, `tip`, `size`, and the calculated `tip_pct` using histograms and KDE plots.

```
plt.figure(figsize=(15, 10))

plt.subplot(2, 2, 1)
sns.histplot(df['total_bill'], kde=True, bins=20, color='skyblue')
plt.title('Distribution of Total Bill')

plt.subplot(2, 2, 2)
sns.histplot(df['tip'], kde=True, bins=20, color='lightcoral')
plt.title('Distribution of Tip')

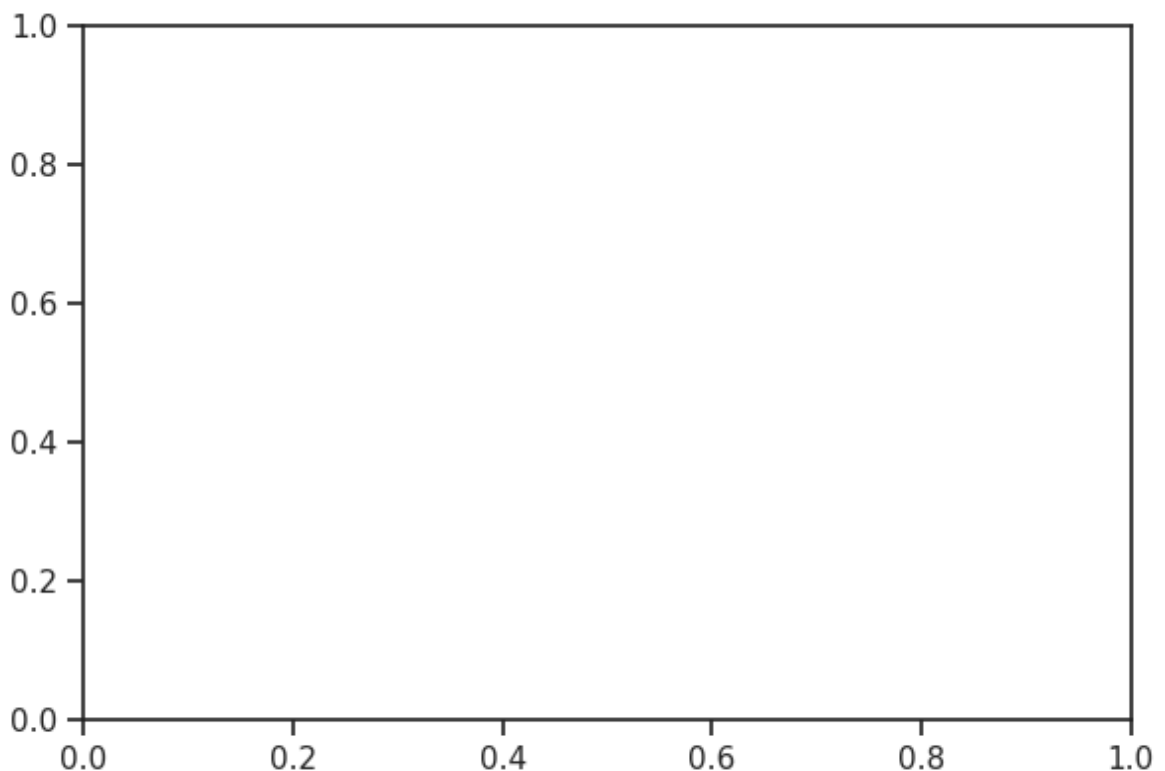
plt.subplot(2, 2, 3)
sns.histplot(df['size'], kde=False, bins=df['size'].nunique(), color='lightgreen')
plt.title('Distribution of Size')

plt.subplot(2, 2, 4)
sns.histplot(df['tip_pct'], kde=True, bins=20, color='gold')
plt.title('Distribution of Tip Percentage')

plt.tight_layout()
plt.show()
```

```
-----
TypeError                                Traceback (most recent call last)
/tmp/ipykernel_2528/2682382771.py in <cell line: 0>()
      2
      3 plt.subplot(2, 2, 1)
----> 4 sns.histplot(df['total_bill'], kde=True, bins=20, color='skyblue')
      5 plt.title('Distribution of Total Bill')
      6
```

TypeError: string indices must be integers, not 'str'



다음 단계: 오류 설명

✓ Multivariate Analysis: Two Variables

Let's explore the relationships between different variables. We'll start with numerical vs. numerical, and then categorical vs. numerical.

```
plt.figure(figsize=(10, 6))
sns.scatterplot(x='total_bill', y='tip', hue='smoker', style='time', data=df, palette='deep')
plt.title('Total Bill vs. Tip by Smoker and Time')
plt.xlabel('Total Bill ($)')
plt.ylabel('Tip ($)')
plt.show()
```

```
-----
TypeError                                Traceback (most recent call last)
/tmp/ipykernel_2528/707480287.py in <cell line: 0>()
      1 plt.figure(figsize=(10, 6))
----> 2 sns.scatterplot(x='total_bill', y='tip', hue='smoker', style='time',
data=df, palette='deep')
      3 plt.title('Total Bill vs. Tip by Smoker and Time')
      4 plt.xlabel('Total Bill ($)')
      5 plt.ylabel('Tip ($)')
```

```
----- 5 frames -----
/usr/local/lib/python3.13/dist-packages/seaborn/_core/data.py in
handle_data_source(data)
    276     elif data is not None and not isinstance(data, Mapping):
    277         err = f"Data source must be a DataFrame or Mapping, not
{type(data)!r}."
--> 278         raise TypeError(err)
    279
    280     return data
```

```
-----
TypeError: Data source must be a DataFrame or Mapping, not <class 'str'>
-----
```

다음 단계: 오류 설명

```
plt.figure(figsize=(12, 6))
sns.boxplot(x='day', y='total_bill', hue='sex', data=df, palette='pastel')
plt.title('Total Bill by Day and Sex')
plt.xlabel('Day of Week')
plt.ylabel('Total Bill ($)')
plt.show()
```

```

-----
TypeError                                Traceback (most recent call last)
/tmp/ipykernel_2528/3389817933.py in <cell line: 0>()
      1 plt.figure(figsize=(12, 6))
----> 2 sns.boxplot(x='day', y='total_bill', hue='sex', data=df, palette='pastel')
      3 plt.title('Total Bill by Day and Sex')
      4 plt.xlabel('Day of Week')
      5 plt.ylabel('Total Bill ($)')

```

5 frames

```

/usr/local/lib/python3.13/dist-packages/seaborn/_core/data.py in
handle_data_source(data)
    276     elif data is not None and not isinstance(data, Mapping):
    277         err = f"Data source must be a DataFrame or Mapping, not
{type(data)!r}."
--> 278         raise TypeError(err)
    279
    280     return data

```

TypeError: Data source must be a DataFrame or Mapping, not <class 'str'>.

다음 단계: 오류 설명

```

plt.figure(figsize=(12, 6))
sns.violinplot(x='time', y='tip_pct', hue='smoker', data=df, palette='muted', inner=
plt.title('Tip Percentage by Time and Smoker')
plt.xlabel('Time of Day')
plt.ylabel('Tip Percentage (%)')
plt.show()

```

```

-----
TypeError                                Traceback (most recent call last)
/tmp/ipykernel_2528/36364953.py in <cell line: 0>()
      1 plt.figure(figsize=(12, 6))
----> 2 sns.violinplot(x='time', y='tip_pct', hue='smoker', data=df,
palette='muted', inner='quartile')
      3 plt.title('Tip Percentage by Time and Smoker')
      4 plt.xlabel('Time of Day')
      5 plt.ylabel('Tip Percentage (%)')

```

5 frames

```

/usr/local/lib/python3.13/dist-packages/seaborn/_core/data.py in
handle_data_source(data)
    276     elif data is not None and not isinstance(data, Mapping):
    277         err = f"Data source must be a DataFrame or Mapping, not
{type(data)!r}."
--> 278         raise TypeError(err)
    279
    280     return data

```

TypeError: Data source must be a DataFrame or Mapping, not <class 'str'>.

다음 단계: 오류 설명

```
df_high_tip = df[df['tip'] / df['total_bill'] >= 0.20]
display(df_high_tip.head())
```

```
-----
TypeError                                Traceback (most recent call last)
/tmp/ipykernel_2528/211870384.py in <cell line: 0>()
----> 1 df_high_tip = df[df['tip'] / df['total_bill'] >= 0.20]
      2 display(df_high_tip.head())
```

TypeError: string indices must be integers, not 'str'

다음 단계: [오류 설명](#)

```
df['tip'] / df['total_bill']
```

	0
0	0.059447
1	0.160542
2	0.166587
3	0.139780
4	0.146808
...	...
239	0.203927
240	0.073584
241	0.088222
242	0.098204
243	0.159744

244 rows × 1 columns

dtype: float64

```
df['tip'] / df['total_bill']*100
```

```

      0
0    5.944673
1   16.054159
2   16.658734
3   13.978041
4   14.680765
...      ...
239 20.392697
240  7.358352
241  8.822232
242  9.820426
243 15.974441

```

244 rows × 1 columns

dtype: float64

```
%whos
```

Variable	Type	Data/Info
df	DataFrame	total_bill tip <...>nWn[244 rows x 7 columns]
sns	module	<module 'seaborn' from '/<...>ges/seaborn/__init__.py'>
tips	DataFrame	total_bill tip <...>nWn[244 rows x 7 columns]

```
df['tip_pct']=(df['tip'] / df['total_bill'])*100
df
```

	total_bill	tip	sex	smoker	day	time	size	tip_pct	
0	16.99	1.01	Female	No	Sun	Dinner	2	5.944673	
1	10.34	1.66	Male	No	Sun	Dinner	3	16.054159	
2	21.01	3.50	Male	No	Sun	Dinner	3	16.658734	
3	23.68	3.31	Male	No	Sun	Dinner	2	13.978041	
4	24.59	3.61	Female	No	Sun	Dinner	4	14.680765	
...	...	...	...	...	...	...	...	...	
239	29.03	5.92	Male	No	Sat	Dinner	3	20.392697	
240	27.18	2.00	Female	Yes	Sat	Dinner	2	7.358352	
241	22.67	2.00	Male	Yes	Sat	Dinner	2	8.822232	
242	17.82	1.75	Male	No	Sat	Dinner	2	9.820426	
243	18.78	3.00	Female	No	Thur	Dinner	2	15.974441	

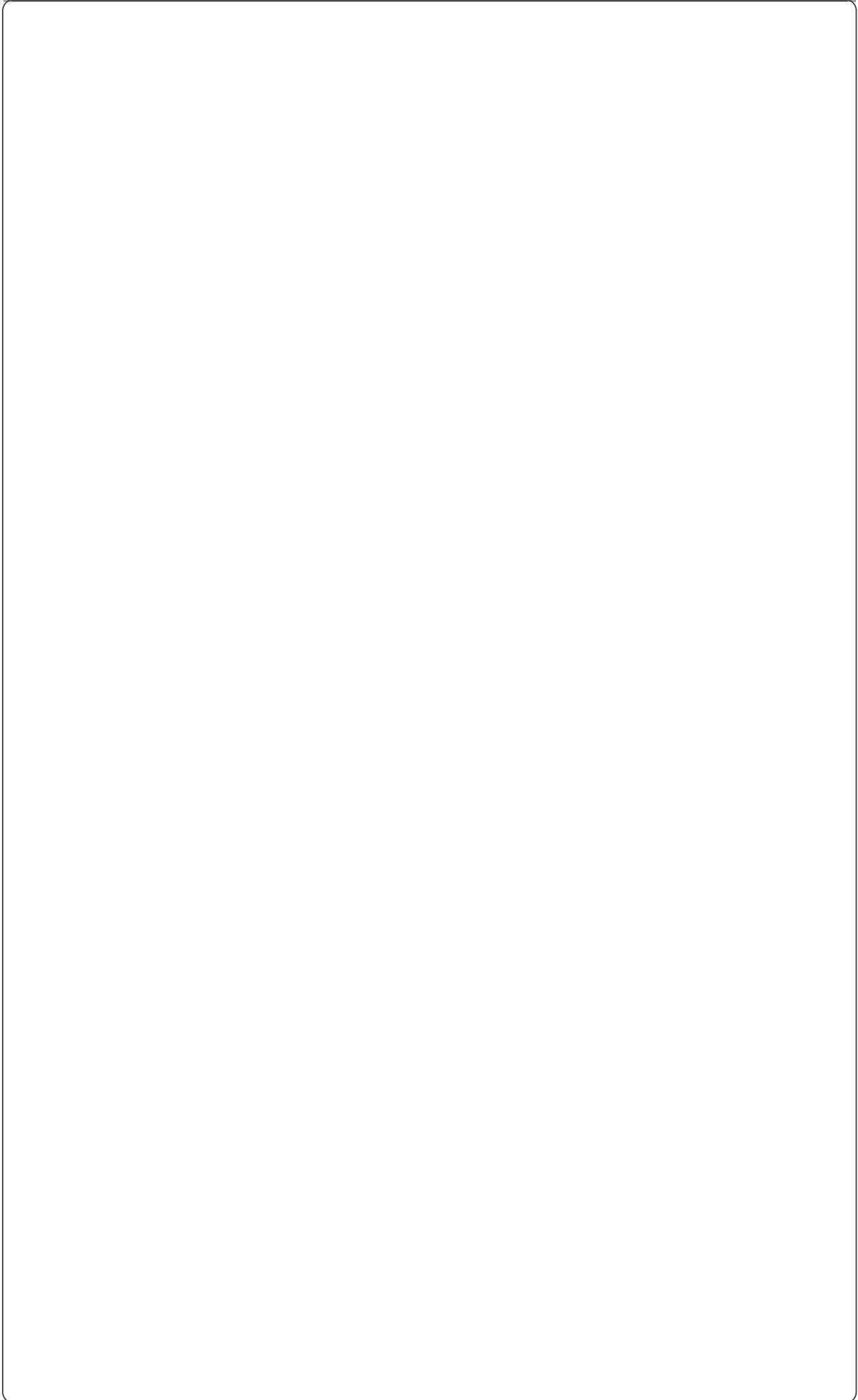
244 rows × 8 columns

다음 단계:

tips 변수로 코드 생성

New interactive sheet

```
df[df['tip_pct']>=20]
```



	total_bill	tip	sex	smoker	day	time	size	tip_pct	
6	8.77	2.00	Male	No	Sun	Dinner	2	22.805017	
9	14.78	3.23	Male	No	Sun	Dinner	2	21.853857	
14	14.83	3.02	Female	No	Sun	Dinner	2	20.364127	
17	16.29	3.71	Male	No	Sun	Dinner	3	22.774708	
18	16.97	3.50	Female	No	Sun	Dinner	3	20.624632	
20	17.92	4.08	Male	No	Sat	Dinner	2	22.767857	

tips									
	total_bill	tip	sex	smoker	day	time	size	tip_pct	
46	22.23	5.00	Male	No	Sun	Dinner	2	22.492128	
51	16.99	1.01	Female	No	Sun	Dinner	2	5.944673	
65	18.29	3.16	Male	Yes	Sat	Dinner	4	20.557682	
1	10.34	1.66	Male	No	Sun	Dinner	3	16.054159	
67	2.07	1.00	Female	Yes	Sat	Dinner	1	22.573300	
2	21.01	3.50	Male	No	Sun	Dinner	3	16.658734	
81	16.00	3.40	Male	No	Fri	Lunch	2	20.408103	
3	23.68	3.31	Male	No	Sun	Dinner	2	13.978041	
67	16.99	1.01	Male	No	Thur	Lunch	2	21.853857	
4	24.59	3.61	Female	No	Sun	Dinner	4	14.680765	
88	24.71	3.85	Male	No	Fri	Lunch	2	23.614020	
...	...	...	...	...	...	...	...	...	
239	29.03	5.92	Male	No	Sat	Dinner	3	20.392697	
100	11.55	2.50	Female	Yes	Fri	Dinner	2	22.020432	
240	27.18	2.00	Female	Yes	Sat	Dinner	2	7.358352	
100	16.99	1.01	Male	No	Sat	Dinner	2	22.614020	
241	22.67	2.00	Male	Yes	Sat	Dinner	2	8.822232	
103	14.51	4.00	Female	Yes	Sat	Dinner	2	21.552401	
242	17.82	1.75	Male	No	Sat	Dinner	2	9.820426	
110	14.00	2.00	Male	No	Sat	Dinner	2	21.428571	
243	18.78	3.00	Female	No	Thur	Dinner	2	15.974441	
115	17.51	3.50	Female	No	Sun	Dinner	2	20.219520	
244 rows × 8 columns									
124	12.48	2.52	Female	No	Thur	Lunch	2	20.192308	
다음 단계: tips 변수로 코드 생성									
139	16.99	1.01	Female	No	Thur	Lunch	2	20.896657	
140	17.47	3.50	Female	No	Thur	Lunch	2	20.034345	

df.head()									
	total_bill	tip	sex	smoker	day	time	size	tip_pct	
172	17.25	5.15	Male	Yes	Sun	Dinner	2	7.1034483	
0	16.99	1.01	Female	No	Sun	Dinner	2	5.944673	13
178	10.34	1.66	Female	No	Sun	Dinner	3	16.054159	67
2	21.01	3.50	Male	No	Sun	Dinner	3	16.658734	45
183	23.68	3.31	Male	No	Sun	Dinner	2	13.978041	3
4	24.59	3.61	Female	No	Sun	Dinner	4	14.680765	64
191	19.81	4.19	Female	Yes	Thur	Lunch	2	21.150934	
다음 단계: df 변수로 코드 생성									
194	16.58	4.00	Male	Yes	Thur	Lunch	2	24.125452	

```
df.groupby('day')['tip'].describe()
```

```

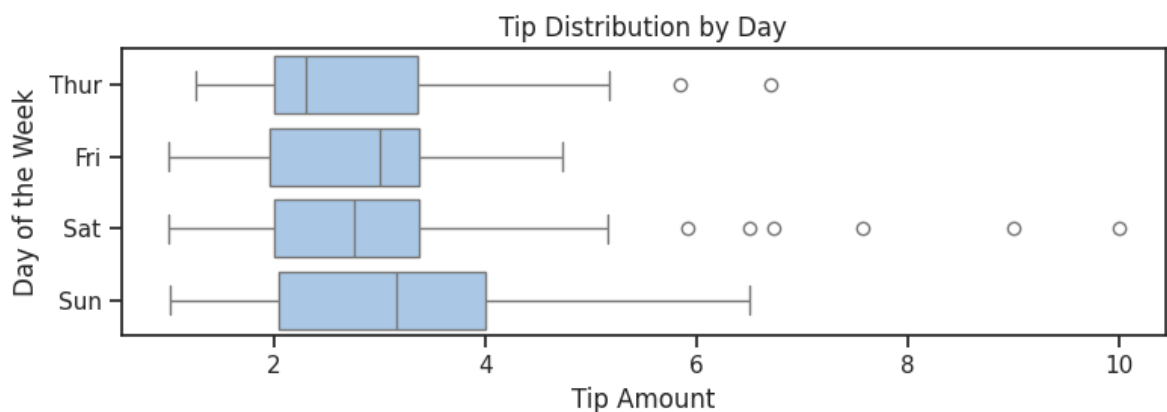
214      28.17    6.30  Female    Yes    Sat    Dinner    3    25.074192
/tmp/ipykernel_2528/3048255687.py:1: FutureWarning: The default of observed=False is
221 df.groupby('day')['tip'].describe()  Yes    Fri    Lunch    2    25.931446
222 count  8.58  1.92    Male    No    Fri    Lunch    75%  22.137762
223 day      13.28  2.72    Male    No    Sat    Dinner    2    20.481928
Thur      62.0  2.771452  1.240223  1.25  2.0000  2.305  3.3625  6.70  366
Fri      19.0  2.0734797  1.019577  1.00  1.9500  3.3650  4.392697
Sat      87.0  2.993103  1.631014  1.00  2.0000  2.750  3.3700  10.00
Sun      76.0  3.255132  1.234880  1.01  2.0375  3.150  4.0000  6.50

```

```

# 이변량
plt.figure(figsize=(8, 3))
sns.boxplot(data=df, x='tip', y='day')
plt.title("Tip Distribution by Day")
plt.xlabel("Tip Amount")
plt.ylabel("Day of the Week")
plt.tight_layout()
plt.show()

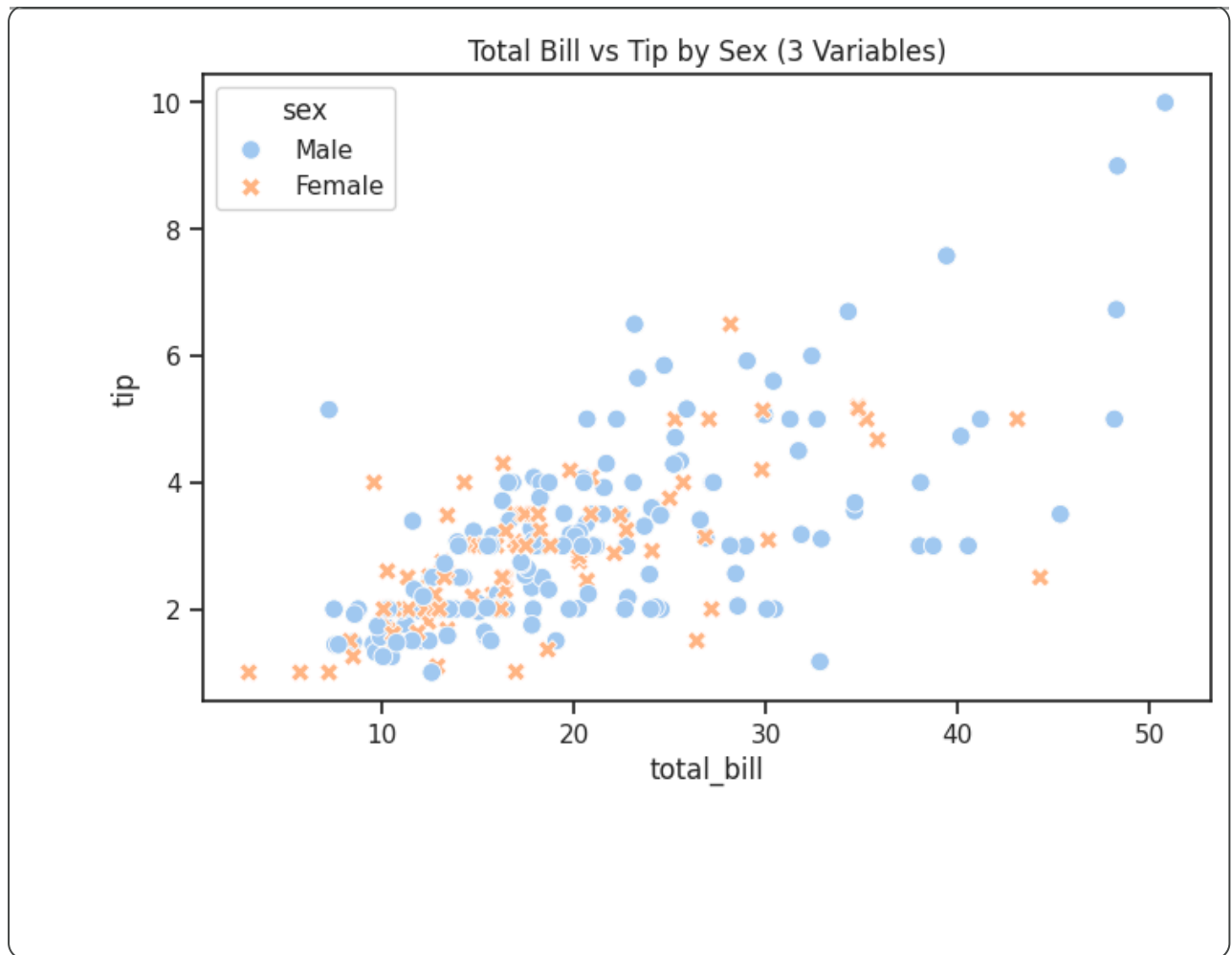
```



```

# 다변량
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x='total_bill', y='tip', hue='sex', style='sex', s=70)
plt.title("Total Bill vs Tip by Sex (3 Variables)")
plt.show()

```



지도를 그려서 median_house_value/population별로 scatter하여 어느 지역의 집값이 비싼지 말하시오(시각화도 하시오).

```
import matplotlib.pyplot as plt
import pandas as pd

file_path='/content/sample_data/california_housing_train.csv'
df=pd.read_csv(file_path)

plt.figure(figsize=(10, 8))

scatter=plt.scatter(
    df['longitude'],
    df['latitude'],
    c=df['median_house_value'],
    cmap='jet',
    s=df['population']/100,
    alpha=0.5,
    edgecolors='none'
)

plt.colorbar(scatter, label='Median House Value ($)')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.title('California Housing Prices Map')
```

```
plt.grid(True, linestyle='--', alpha=0.5)
```

